

A Multi-Source Machine Learning Framework for Real-Time Road Hazard Detection Using Traffic Cameras and Atmospheric Sensor Data

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Introduction

Weather-related road hazards such as snow, ice, heavy rain, and fog significantly increase accident risk and traffic congestion. In Central Texas, the “Flash Flood Alley” geography combined with recent extreme winter weather events (such as 2021 and 2023 ice storms) have highlighted a critical gap in municipal preparedness.

This research proposes a **Multi-Sensor Data Fusion (MSDF)** framework that transcends the limitations of traditional, visual-only monitoring. While standard TxDOT cameras fail to detect "invisible" hazards like black ice or high-velocity wind, this system integrates visual imagery with **NOAA atmospheric telemetry** (dew point, pressure) to generate a predictive "surplus" of information.

The study introduces a region-specific **ETL (Extract, Transform, Load) pipeline** tailored to the **I-35 corridor** (Austin to San Marcos), synchronizing high-frequency camera data with localized weather readings. This approach solves the information-lag inherent in current systems, enabling real-time prediction of hazardous road conditions before they manifest physically on the asphalt.

Research Focus:

This research aims to bridge the "visual-only" gap in intelligent transportation by developing an automated Multi-Sensor Data Fusion (MSDF) system to classify road hazards (safe, wet, snowy, icy, storm-risk). While standard TxDOT cameras struggle with invisible threats like black ice or night-time hazards, this study integrates **NOAA atmospheric-telemetry** specifically barometric pressure, humidity, and temperature to achieve a predictive surplus of information.

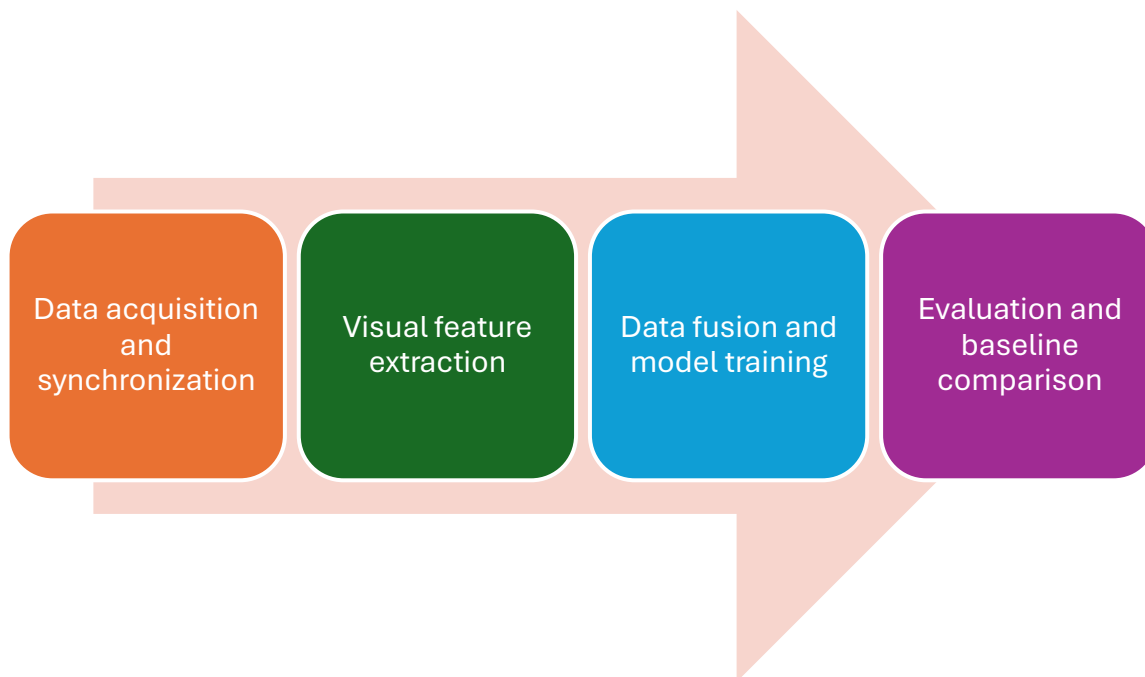
Advised by Dr. Castro-Hernandez, this project involves the end-to-end design of a region-specific **Python-based ETL pipeline** tailored to the I-35 corridor. Key milestones include:

- **Quantifying the Fusion Advantage:** Training a **Random Forest classifier** to compare fused data performance against a camera-only baseline, specifically measuring F1-score improvements.
- **Feature Importance Ranking:** Identifying which specific variables (e.g., dew point vs. visual texture) serves as the strongest predictors for "invisible" hazards.

- **Reliability Evaluation:** Determining the specific thresholds where atmospheric data supersedes visual data, such as during heavy fog or low-light conditions, to provide a roadmap for more resilient municipal safety dashboards.

Methodology:

The proposed system will be implemented using modular Python architecture, utilizing OpenCV for computer vision, NumPy/Pandas for data processing, and Scikit-learn for machine learning. The methodology follows a four-state pipeline:



Data acquisition and synchronization

The first stage establishes a custom pipeline collecting TxDOT camera images every 5–10 minutes and NOAA atmospheric telemetry (example, pressure, visibility) via REST APIs. To ensure spatial-temporal integrity, each camera coordinate is paired with the nearest active weather station.

Visual feature extraction

Raw images will be pre-processed to remove noise and standardized to a fixed resolution. Using OpenCV, I will extract three primary descriptors:

Data fusion and model training

Visual indicators are fused with atmospheric data to create high-dimensional feature vectors (~30k–50k samples). Hazard labels (e.g. safe, icy, storm-risk) are assigned based on environmental thresholds like temperature ($<0^{\circ}\text{C}$) and visibility metrics. A Random Forest classifier is employed for its ability to handle non-linear relationships and robustness against outliers.

Evaluation and Baseline Comparison

To quantify the fusion advantage, I will compare a Camera-Only baseline against the fused model using 5-fold cross-validation. Performance will be evaluated primarily via F1-score, alongside accuracy, precision, and recall metrics.

FUTURE WORK

Future work prioritizes enhancing model accuracy through Deep Learning (CNNs) and expanding the framework to new regions and environmental threats. The goal is a scalable, real-time monitoring dashboard for transportation agencies and drivers. To foster innovation, the project will be open-sourced, enabling community-driven deployment across Central Texas and beyond.

CONCLUSION

Addressing critical infrastructure gaps in Central Texas, this research deploys a **Multi-Sensor Data Fusion (MSDF)** framework to detect hazards invisible to standard cameras. By using **TxDOT visual streams** with **NOAA atmospheric-telemetry** via a custom **ETL pipeline** and **Random Forest classification**, the project delivers a robust, real-time decision-support system. The study ensures scalability and transparency by **open sourcing** its code, directly contributing to a safer, more resilient transportation network for the Texas State community.

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